Service & Travel Proxy Model Specification

1. Overview

This document describes the **service model** used in the EMS Readiness Optimization simulation. The model has two components:

Component	Purpose	Module
Travel-time proxy	Estimate response time from firehouse to incident	`service/travel_time.py`
Service-time distribution	Sample on-scene + return-to-available duration	`service/service_time.py`

Both feed into the SimPy discrete-event simulation (Phase 3).

---

2. Travel-Time Proxy

2.1 Approach

We approximate travel time with:

$$
t_{\text{travel}} = \frac{d_{\text{haversine}}}{v_{\text{avg}}}
$$

where:

- $d_{\text{haversine}}$ is the great-circle (Haversine) distance between the firehouse and the incident/precinct centroid.
- v_{avg} is a configurable average EMS speed (default: **20 mph**).

2.2 Justification

Consideration	Detail
Why Haversine?	Simple, reproducible, requires only lat/lon. Manhattan's grid layout means Haversine consistently underestimates road distance by a "detour factor" of ~1.3–1.4, but the relative ordering of firehouse–precinct pairs is preserved.
Why not road-network routing?	Per the project charter, full network routing is **out of scope**. A distance-based proxy is standard in EMS operations research (Goldberg 2004, Ingolfsson et al. 2008).
Why 20 mph?	NYC DOT reports average Manhattan traffic speeds of 12–25 mph. EMS vehicles with lights & sirens travel faster than ambient traffic. 20 mph is the midpoint used in prior NYC EMS studies.

2.3 Time-of-Day Variation (Optional)

Speed multipliers can be applied by hour of day:

Period	Hours	Speed Factor	Effective Speed
Overnight	00–06	1.30	26.0 mph
AM Peak	06–10	0.75	15.0 mph
Midday	10–16	0.90	18.0 mph
PM Peak	16–20	0.70	14.0 mph
Evening	20–24	1.10	22.0 mph

These factors are derived from NYC DOT speed monitoring data and can be toggled on/off in `configs/service.yaml`.

2.4 Distance Matrix

A pre-computed distance matrix is stored at:

```
data/processed/distance_matrix_firehouse_precinct.csv
```

- **Rows**: 48 Manhattan firehouses (by facility name).
- **Columns**: 30 Manhattan precincts (by precinct number).
- **Values**: Haversine distance in miles.

3. Service-Time Distribution

3.1 Distribution Choice: LogNormal

We model on-scene service time with a **LogNormal** distribution.

Why LogNormal over Exponential?

Criterion	LogNormal	Exponential
Positivity	✅ Always > 0	✅ Always > 0
Skewness	✅ Right-skewed (matches EMS data)	✅ Right-skewed
Mode ≠ 0	✅ Has a meaningful mode	❌ Mode = 0
Coefficient of variation	✅ Flexible (CV < 1 possible)	❌ Fixed at CV = 1
Empirical fit	✅ Better fit in EMS literature	❌ Over-predicts short calls

3.2 Parameters

Parameter	Value	Rationale
Mean	25 minutes	Mid-range of 20–30 min typical in literature
Std Dev	10 minutes	CV ≈ 0.4, consistent with EMS data

These are converted internally to LogNormal (μ, σ) via moment-matching:

$$
\sigma^2 = \ln\!\left(1 + \frac{\text{Var}}{\text{Mean}^2}\right), \quad
\mu = \ln(\text{Mean}) - \frac{\sigma^2}{2}
$$

3.3 What Service Time Covers

The sampled service time represents the **on-scene + return-to-available** duration. The full call timeline is:

1. **Dispatch delay** — fixed 1.5 min (configurable)
2. **Travel to scene** — from travel-time proxy
3. **On-scene service** — from service-time distribution ← this component
4. **Return to available** — included in service-time draw

4. Limitations & Assumptions

#	Assumption	Risk	Mitigation
1	Haversine underestimates road distance	Response times may be optimistic	Sensitivity analysis with speed ± 25%
2	Average speed is constant within a TOD period	Ignores micro-level congestion	TOD factors capture macro patterns
3	Service time is i.i.d. across calls	Call severity varies	LogNormal captures heterogeneity in tail
4	No mutual-aid or cross-borough dispatch	Some calls may be served from outside Manhattan	Conservative (overstates Manhattan-only demand)
5	Precinct centroids represent demand locations	Demand is spread across precinct	Centroid is acceptable for strategic-level analysis

---

5. Configuration

All parameters are centralized in:

- `configs/service.yaml` — speed, service-time distribution, dispatch delay
- `configs/demand.yaml` — base rate, lambda table paths, simulation defaults

---

6. References

1. Goldberg, J. B. (2004). Operations Research Models for the Deployment of Emergency Services Vehicles. EMS Management Journal.
2. Ingolfsson, A., Budge, S., & Erkut, E. (2008). Optimal ambulance location with random delays and travel times. Health Care Management Science, 11(3), 262–274.
3. NYC DOT. NYC Mobility Report — average traffic speeds in Manhattan.
4. Lewis, P. A. W., & Shedler, G. S. (1979). Simulation of nonhomogeneous Poisson processes by thinning. Naval Research Logistics Quarterly, 26(3), 403–413.